

Yibo Tu

Name	Activity
Yibo Tu	1.Decompile APK files and use AI to identify code vulnerabilities. 2.Analyzed the security issue in Login.generateSessionToken() using java.util.Random. 3.Explained why this method is unsuitable for generating session tokens. 4.Described a realistic attacker model and capabilities. 5.Created a step-by-step attack scenario. 6.Compiled Task 4 into an AI-log format for submission.
Weihaio Cheng	1.Completed Task 2 2.Finished the relevant sections of Task 2 in both the report and the AI_Log 3.Contributed to organizing and structuring the report format
Yucheng Zhao	1.Installed and configured JADX, then successfully decompiled the provided APK 2.Used AI assistance to identify suitable APK decompilation tools on macOS and selected JADX 3.Identified the package name, main activity, AndroidManifest.xml, and authentication-related classes 4.Collecting supporting evidence, including screenshots of the decompiled login-related code 5.Explained what an APK file is and why decompilation is necessary for security auditing 6.Drafted and refined the Task 1 section and prepared the corresponding AI log entry for submission
Xu He	1.Responsible for identifying vulnerabilities in the APK 2.Completed the Task 3 analysis and prepared the corresponding report and AI log
Yuchen Zhou	1.Defined the attack model, including different attacker types, their capabilities, and access levels. 2.Presented a four-step impersonation attack scenario demonstrating how the vulnerability can be exploited. 3.Explained how the vulnerability arises in the code and the conditions required for exploitation. 4.Discussed why the vulnerability is unsuitable for its intended security purpose.

Weihaio Cheng

Name	Activity
Yibo Tu	1.APK Decompilation and AI-Assisted Vulnerability Detection 2.Analyzed the APK vulnerability 3.Completed the attacker section corresponding to the report
Weihaio Cheng	1.Complete the Task2 2.Completed the relevant part in task 2 of the report and AI_Log 3.Assisted in constructing the report format
Yucheng Zhao	1.Write and refine the content for Task 1 and the corresponding AI_log 2.Decompile the APK in the early stage of the project
Xu He	1.Responsible for identifying vulnerabilities in the APK 2.Completed the Task 3 analysis, and prepared the corresponding report and AI log
Yuchen Zhou	1.Clearly explained the cause of the vulnerability, the location of the code, and the conditions under which it can be exploited. 2.Build attack models, define attacker types 3.Analyzed why the vulnerability does not apply to its expected security function.

Yucheng Zhao

Name	Activity
Yibo Tu	1.Contributed to the overall vulnerability assessment of the decompiled APK and helped identify security issues in the application. 2.Developed the attacker model, created the attack scenario, and organized the Task 4 AI-log for submission.
Weihaio Cheng	1.Complete the Task2 2.Completed the relevant part in task 2 of the report and AI_Log 3.Assisted in constructing the report format
Yucheng Zhao	1.Used AI assistance to identify suitable APK decompilation tools on macOS and selected JADX. 2.Installed and configured JADX, then successfully decompiled the provided APK. 3.Identified the package name, main activity, AndroidManifest.xml, and

	authentication-related classes. 4.Explained what an APK file is and why decompilation is necessary in security auditing. 5.Collected supporting evidence, including screenshots of the decompiled Login-related code. 6.Drafted and refined the Task 1 section and prepared the related AI-log entry for submission.
Xu He	1.Responsible for identifying vulnerabilities in the APK 2.Completed the Task 3 analysis, and prepared the corresponding report and AI log
Yuchen Zhou	1.Clearly explained the cause of the vulnerability, the location of the code, and the conditions under which it can be exploited. 2.Build attack models, define attacker types 3.Analyzed why the vulnerability does not apply to its expected security function.

Xu He

Name	Activity
Yibo Tu	1.Contributed to the overall vulnerability assessment of the decompiled APK and helped identify security issues in the application. 2.Developed the attacker model, created the attack scenario, and organized the Task 4 AI-log for submission.
Weihaio Cheng	1.Mainly contributed to the completion of Task 2. 2.Wrote the relevant Task 2 section of the report and prepared the corresponding AI-log content. 3.Assisted in structuring and formatting the report to improve consistency and readability.
Yucheng Zhao	1.Used AI assistance to identify a suitable APK decompilation tool for macOS, selected JADX, and completed its installation and configuration. 2.Successfully decompiled the provided APK and identified key application components, including the package name, main activity, 3.AndroidManifest.xml, and authentication-related classes. Drafted and refined the Task 1 section, and prepared supporting screenshots and the related AI-log entry.
Xu He	1.Reviewed the decompiled APK source code and identified two randomness-related locations: MainActivity.randomNumberGenerator() and Login.generateSessionToken(). 2.Analyzed the security role of both values and determined that only Login.generateSessionToken() was security-sensitive because it generated a session token. 3.Verified the code flow showing that the generated token was stored through createSession() in KEY_SESSION_TOKEN after successful login. 4.Identified Login.generateSessionToken() as the core Task 3 vulnerability because it used java.util.Random to generate a

	security-sensitive session token. 5.Wrote and refined the Task 3 report content, including the static analysis results, security role analysis, and core vulnerability determination. 6.Prepared pocs materials by collecting the key code evidence and organizing the screenshots used to demonstrate the random-value locations and the session token generation flow. 7.Prepared the Task 3 AI log by recording the main prompts, outcomes, and rubric-based mock Q&A related to the randomness vulnerability analysis.
Yuchen Zhou	1.Clearly explained the identified vulnerabilities, including how they arise in the code and the conditions required for successful exploitation. 2.Explained why the vulnerability is inappropriate for its intended security purpose. 3.Defined attacker types and capabilities, and developed a detailed four-step impersonation attack scenario.

Yuchen Zhou

Name	Activity
Yibo Tu	1.Analyzed key vulnerabilities and explained their security risks. 2.Defined a realistic attacker model that operates without root privileges. 3.Established attacker capabilities, including dynamic sample collection and offline seed cracking. 4.Constructed a 4-step scenario demonstrating session hijacking through token prediction and impersonation
Weihaio Cheng	1.Completed Task 2. 2.Wrote and finalized the relevant sections of the report and AI log. 3.Contributed to structuring and formatting the report for better consistency and readability.
Yucheng Zhao	1.Used AI assistance to select JADX as a suitable APK decompilation tool. 2.Installed and configured JADX, and successfully decompiled the provided APK. 3.Identified key components, including the package name, main activity, AndroidManifest.xml, and authentication-related classes. 4.Explained what an APK file is and the importance of decompilation in security auditing. 5.Collecting supporting evidence, such as screenshots of decompiled login-related code. 6.Drafted and refined the Task 1 report and completed the corresponding AI log entry.
Xu He	1.Identified and analyzed vulnerabilities in the decompiled APK. 2.Detected a randomness-related vulnerability in session token generation.

	3.Analyzed the code flow and confirmed the security impact of the vulnerability. 4.Collected supporting evidence, including key code snippets and screenshots. 5.Completed the Task 3 analysis and finalized the report and AI log.
Yuchen Zhou	1.Provided a clear explanation of the identified vulnerabilities, including how they arise in the code and the specific conditions required for successful exploitation. 2.Explained why the vulnerability is unsuitable for its intended security role. 3.Defined different types of attackers and outlined their capabilities, assumptions, and access levels within the attack model. 4.Developed a detailed four-step impersonation attack scenario.